Vector Spaces and Advanced Linear Algebra Concepts for Machine Learning and Deep Learning

Akshay Redekar

January 25, 2026

Contents

1 Vector Spaces **3**
1.1 Introduction and Motivation . . . 3
1.2 Definition of Vector Spaces . . . 3
1.3 Standard Examples of Vector Spaces . . . 4
1.4 Properties of Vector Spaces . . . 5

2 Subspaces **5**
2.1 Definition and Properties . . . 5
2.2 Operations on Subspaces . . . 7

3 Span and Linear Combinations **8**
3.1 Linear Combinations . . . 8
3.2 Span . . . 9
3.3 Spanning Sets and Generating Sets . . . 10

4 Linear Independence and Dependence **10**
4.1 Definitions . . . 10
4.2 Testing Linear Independence . . . 11
4.3 Properties of Linear Independence . . . 12

5 Basis and Dimension **12**
5.1 Definition of Basis . . . 12
5.2 Standard Bases . . . 13
5.3 Dimension . . . 14
5.4 Subspace Dimensions . . . 15

6 Change of Basis **16**
6.1 Coordinate Vectors . . . 16
6.2 Change of Basis Matrix . . . 16
6.3 Algorithm for Change of Basis . . . 17

7 Applications in Machine Learning and Deep Learning **18**
7.1 Feature Spaces and Data Representation 18
7.2 Principal Component Analysis (PCA) 18
7.3 Neural Network Weight Spaces 19
7.4 Kernel Methods and Infinite Dimensions 19
7.5 Word Embeddings 20

8 Worked Examples **21**
8.1 Complete Example 1: Verifying a Basis 21
8.2 Complete Example 2: Finding Coordinates 21
8.3 Complete Example 3: Dimension of Subspace 22

9 Assessment Questions **23**
9.1 Level 1: Conceptual Understanding 23
9.2 Level 2: Computational 24
9.3 Level 3: Advanced and Proofs 24

10 Solutions to Selected Problems **25**
10.1 Solution to Q11 25
10.2 Solution to Q12 26
10.3 Solution to Q13 26
10.4 Solution to Q21 (Proof) 27

11 Practice Problems **28**
11.1 Computational Practice 28
11.2 Conceptual Practice 28
11.3 ML/DL Application Practice 29

12 Summary and Key Takeaways **29**

13 Further Reading and Resources **31**
13.1 Textbooks 31
13.2 Online Resources 31
13.3 Research Papers and Articles 32
13.4 Interactive Tools 32

14 Appendix: Python Implementations **33**
14.1 Testing Linear Independence 33
14.2 Finding Basis for Null Space 33
14.3 Coordinate Vector Computation 34
14.4 Change of Basis Matrix 35
14.5 Dimension of Subspace Intersection 36

15 Conclusion **37**

1 Vector Spaces

1.1 Introduction and Motivation

Vector spaces are fundamental structures in mathematics that generalize the notion of vectors in $\mathbb{R}^n$. They provide the framework for understanding linear algebra, functional analysis, and are essential in machine learning for feature spaces, model representations, and optimization landscapes.

Why Vector Spaces in ML/DL?

Critical Applications:

1. **Feature Spaces:** Input data lives in vector spaces (e.g., $\mathbb{R}^d$ for d features)
2. **Model Parameters:** Weights and biases form vector spaces
3. **Function Spaces:** Neural networks approximate functions in infinite-dimensional spaces
4. **Kernel Methods:** Transform data to higher-dimensional vector spaces
5. **Optimization:** Gradient descent operates in parameter vector spaces
6. **Dimensionality Reduction:** Finding optimal subspaces (PCA, autoencoders)
7. **Embeddings:** Word/image embeddings are vectors in learned spaces

1.2 Definition of Vector Spaces

Definition 1.1 (Vector Space)**.** A vector space V over a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$) is a non-empty set equipped with two operations:

1. **Vector Addition:** $+ : V \times V \to V$
2. **Scalar Multiplication:** $\cdot : \mathbb{F} \times V \to V$

These operations must satisfy the following axioms for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\alpha, \beta \in \mathbb{F}$:
Axioms for Vector Addition:

$$\text{A1. Closure:} \quad \mathbf{u} + \mathbf{v} \in V \tag{1}$$
$$\text{A2. Commutativity:} \quad \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u} \tag{2}$$
$$\text{A3. Associativity:} \quad (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}) \tag{3}$$
$$\text{A4. Identity:} \quad \exists \mathbf{0} \in V \text{ s.t. } \mathbf{v} + \mathbf{0} = \mathbf{v} \tag{4}$$
$$\text{A5. Inverse:} \quad \forall \mathbf{v} \in V, \exists (-\mathbf{v}) \in V \text{ s.t. } \mathbf{v} + (-\mathbf{v}) = \mathbf{0} \tag{5}$$

Axioms for Scalar Multiplication:

M1. Closure: $\alpha\mathbf{v} \in V$ (6)

M2. Compatibility: $\alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v}$ (7)

M3. Identity: $1 \cdot \mathbf{v} = \mathbf{v}$ (8)

M4. Distributivity (scalar): $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$ (9)

M5. Distributivity (vector): $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$ (10)

1.3 Standard Examples of Vector Spaces

Example 1.1 (Euclidean Space $\mathbb{R}^n$)**.** The set of all n-tuples of real numbers:

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{R} \right\} \tag{11}$$

Operations:

$$\text{Addition: } \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix} \tag{12}$$

$$\text{Scalar mult: } \alpha \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix} \tag{13}$$

ML Application: Feature vectors in classification, regression.

Example 1.2 (Matrix Space $\mathbb{R}^{m\times n}$)**.** The set of all $m \times n$ matrices with real entries:

$$\mathbb{R}^{m\times n} = \{A : A \text{ is } m \times n \text{ matrix with entries in } \mathbb{R}\} \tag{14}$$

Operations:

$$(A + B)_{ij} = A_{ij} + B_{ij} \tag{15}$$

$$(\alpha A)_{ij} = \alpha A_{ij} \tag{16}$$

ML Application: Weight matrices in neural networks, convolutional filters.

Example 1.3 (Polynomial Space P_n)**.** The set of all polynomials of degree at most n:

$$P_n = \{a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n : a_i \in \mathbb{R}\} \tag{17}$$

Operations:

$$p(x) + q(x) = \sum_{i=0}^{n} (a_i + b_i) x^i \tag{18}$$

$$\alpha p(x) = \sum_{i=0}^{n} (\alpha a_i) x^i \tag{19}$$

Dimension: $\dim(P_n) = n + 1$
ML Application: Polynomial regression, basis functions.

Example 1.4 (Function Space $C[a,b]$. $]These of all continuous real-valued functions on interval$[a,b]: $\{f : [a,b] \to \mathbb{R} \mid f \text{ is continuous}\}(20)$

Operations:

$$(f+g)(x) = f(x) + g(x) \tag{21}$$
$$(\alpha f)(x) = \alpha f(x) \tag{22}$$

ML Application: Neural networks as function approximators, kernel methods.

Example 1.5 (Sequence Space ℓ^2)**.** The set of all square-summable infinite sequences:

$$\ell^2 = \left\{ (x_1, x_2, x_3, \ldots) : \sum_{i=1}^{\infty} |x_i|^2 < \infty \right\} \tag{23}$$

ML Application: Infinite-dimensional feature spaces in kernel methods.

1.4 Properties of Vector Spaces

Theorem 1.1 (Basic Properties)**.** Let V be a vector space over field $\mathbb{F}$. For any $\mathbf{v} \in V$ and $\alpha \in \mathbb{F}$:

1. **Uniqueness of zero:** The zero vector $\mathbf{0}$ is unique
2. **Uniqueness of inverse:** For each $\mathbf{v}$, the additive inverse $-\mathbf{v}$ is unique
3. **Zero multiplication:** $0 \cdot \mathbf{v} = \mathbf{0}$ and $\alpha \cdot \mathbf{0} = \mathbf{0}$
4. **Negative multiplication:** $(-1) \cdot \mathbf{v} = -\mathbf{v}$
5. **Cancellation:** If $\alpha\mathbf{v} = \mathbf{0}$, then $\alpha = 0$ or $\mathbf{v} = \mathbf{0}$

Proof of Property 3. We prove $0 \cdot \mathbf{v} = \mathbf{0}$:

$$0 \cdot \mathbf{v} = (0+0) \cdot \mathbf{v} \quad \text{(property of 0)} \tag{24}$$
$$= 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v} \quad \text{(distributivity)} \tag{25}$$
$$0 \cdot \mathbf{v} + (-(0 \cdot \mathbf{v})) = (0 \cdot \mathbf{v} + 0 \cdot \mathbf{v}) + (-(0 \cdot \mathbf{v})) \tag{26}$$
$$\mathbf{0} = 0 \cdot \mathbf{v} + (0 \cdot \mathbf{v} + (-(0 \cdot \mathbf{v}))) \tag{27}$$
$$\mathbf{0} = 0 \cdot \mathbf{v} + \mathbf{0} \tag{28}$$
$$\mathbf{0} = 0 \cdot \mathbf{v} \tag{29}$$

□

2 Subspaces

2.1 Definition and Properties

Definition 2.1 (Subspace)**.** A subset W of a vector space V is called a **subspace** if W is itself a vector space under the same operations as V.

Equivalent condition (Subspace Test): $W \subseteq V$ is a subspace if and only if:

1. $\mathbf{0} \in W$ (contains zero vector)
2. $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$ (closed under addition)
3. $\alpha \in \mathbb{F}, \mathbf{v} \in W \Rightarrow \alpha\mathbf{v} \in W$ (closed under scalar multiplication)

Compact form: W is a subspace if for all $\alpha, \beta \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in W$:

$$\alpha\mathbf{u} + \beta\mathbf{v} \in W \tag{30}$$

Example 2.1 (Subspaces of $\mathbb{R}^3$)**.** Let $V = \mathbb{R}^3$. The following are subspaces:

1. Trivial subspace: $W_0 = \{\mathbf{0}\}$

2. Lines through origin:

$$W_1 = \left\{ t \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\} \tag{31}$$

Verification:

- Contains $\mathbf{0}$: $t = 0 \Rightarrow \mathbf{0} \in W_1$
- Closed under addition: $t_1\mathbf{v} + t_2\mathbf{v} = (t_1 + t_2)\mathbf{v} \in W_1$
- Closed under scalar mult: $\alpha(t\mathbf{v}) = (\alpha t)\mathbf{v} \in W_1$

3. Planes through origin:

$$W_2 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y - z = 0 \right\} \tag{32}$$

Verification: Check if $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ satisfy $u_1 + u_2 - u_3 = 0$ and $v_1 + v_2 - v_3 = 0$.

For $\alpha\mathbf{u} + \beta\mathbf{v}$:

$$(\alpha u_1 + \beta v_1) + (\alpha u_2 + \beta v_2) - (\alpha u_3 + \beta v_3) \tag{33}$$
$$= \alpha(u_1 + u_2 - u_3) + \beta(v_1 + v_2 - v_3) \tag{34}$$
$$= \alpha(0) + \beta(0) = 0 \tag{35}$$

4. The whole space: $W_3 = \mathbb{R}^3$

Non-example: A plane not through origin is NOT a subspace.

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y - z = 1 \right\} \quad \text{(NOT a subspace)} \tag{36}$$

Does not contain $\mathbf{0}$ since $0 + 0 - 0 = 0 \neq 1$.

Example 2.2 (Column Space and Null Space)**.** For matrix $A \in \mathbb{R}^{m\times n}$:

1. Column Space:

$$\mathcal{C}(A) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\} \tag{37}$$

This is a subspace of $\mathbb{R}^m$ (the codomain).

2. Null Space (Kernel):

$$\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\} \tag{38}$$

This is a subspace of $\mathbb{R}^n$ (the domain).

Example:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \tag{39}$$

Column space:

$$\mathcal{C}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^3 \tag{40}$$

Null space: Solve $A\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{41}$$

This gives $x_1 + 2x_2 + x_3 = 0$, so:

$$\mathcal{N}(A) = \text{span}\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^3 \tag{42}$$

2.2 Operations on Subspaces

Theorem 2.1 (Intersection of Subspaces)**.** If W_1 and W_2 are subspaces of V, then $W_1 \cap W_2$ is also a subspace of V.

Proof. We verify the subspace conditions:

1. $\mathbf{0} \in W_1$ and $\mathbf{0} \in W_2 \Rightarrow \mathbf{0} \in W_1 \cap W_2$
2. If $\mathbf{u}, \mathbf{v} \in W_1 \cap W_2$, then:
   - $\mathbf{u}, \mathbf{v} \in W_1 \Rightarrow \mathbf{u} + \mathbf{v} \in W_1$
   - $\mathbf{u}, \mathbf{v} \in W_2 \Rightarrow \mathbf{u} + \mathbf{v} \in W_2$
   - Therefore, $\mathbf{u} + \mathbf{v} \in W_1 \cap W_2$
3. Similar argument for scalar multiplication

□

Definition 2.2 (Sum of Subspaces)**.** If W_1 and W_2 are subspaces of V, their sum is:

$$W_1 + W_2 = \{\mathbf{w}_1 + \mathbf{w}_2 : \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2\} \tag{43}$$

This is also a subspace of V.

Definition 2.3 (Direct Sum)**.** If $W_1 \cap W_2 = \{\mathbf{0}\}$, we write:

$$V = W_1 \oplus W_2 \tag{44}$$

and say V is the **direct sum** of W_1 and W_2.

Property: Every $\mathbf{v} \in V$ has a unique decomposition $\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2$ with $\mathbf{w}_1 \in W_1$, $\mathbf{w}_2 \in W_2$.

3 Span and Linear Combinations

3.1 Linear Combinations

Definition 3.1 (Linear Combination)**.** A vector $\mathbf{v}$ is a **linear combination** of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that:

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = \sum_{i=1}^{n} \alpha_i\mathbf{v}_i \tag{45}$$

The scalars α_i are called **coefficients**.

Example 3.1 (Linear Combination in $\mathbb{R}^3$)**.** Express $\mathbf{v} = \begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix}$ as a linear combination of $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$.

Setup:

$$\alpha_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix} \tag{46}$$

System of equations:

$$\alpha_1 = 5 \tag{47}$$
$$2\alpha_1 + \alpha_2 = 7 \tag{48}$$
$$\alpha_1 - \alpha_2 = 3 \tag{49}$$

Solution: From equation (1): $\alpha_1 = 5$
Substitute into equation (2): $2(5) + \alpha_2 = 7 \Rightarrow \alpha_2 = -3$
Check with equation (3): $5 - (-3) = 8 \neq 3$
Conclusion: $\mathbf{v}$ cannot be expressed as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

3.2 Span

Definition 3.2 (Span)**.** The **span** of a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is the set of all linear combinations:

$$\text{span}(S) = \text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_n\} = \left\{ \sum_{i=1}^{n} \alpha_i \mathbf{v}_i : \alpha_i \in \mathbb{F} \right\} \tag{50}$$

Key property: $\text{span}(S)$ is the smallest subspace containing all vectors in S.

Theorem 3.1 (Span is a Subspace)**.** For any set of vectors $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ in vector space V, $\text{span}(S)$ is a subspace of V.

Proof. We verify the subspace conditions:

1. **Contains zero:** Take all $\alpha_i = 0$:
$$\mathbf{0} = \sum_{i=1}^{n} 0 \cdot \mathbf{v}_i \in \text{span}(S) \tag{51}$$
2. **Closed under addition:** Let $\mathbf{u} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$ and $\mathbf{w} = \sum_{i=1}^{n} \beta_i \mathbf{v}_i$:
$$\mathbf{u} + \mathbf{w} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i + \sum_{i=1}^{n} \beta_i \mathbf{v}_i = \sum_{i=1}^{n} (\alpha_i + \beta_i)\mathbf{v}_i \in \text{span}(S) \tag{52}$$
3. **Closed under scalar multiplication:** Let $\mathbf{u} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$ and $c \in \mathbb{F}$:
$$c\mathbf{u} = c \sum_{i=1}^{n} \alpha_i \mathbf{v}_i = \sum_{i=1}^{n} (c\alpha_i)\mathbf{v}_i \in \text{span}(S) \tag{53}$$

□

Example 3.2 (Span in $\mathbb{R}^3$)**. Example 1:** Single vector

$$\text{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\} = \left\{ t \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\} \tag{54}$$

This is a line through the origin.

Example 2: Two non-collinear vectors

$$\text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} s \\ t \\ 0 \end{bmatrix} : s, t \in \mathbb{R} \right\} \tag{55}$$

This is the xy-plane.

Example 3: Three non-coplanar vectors

$$\text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^3 \tag{56}$$

This spans all of $\mathbb{R}^3$.

Example 3.3 (Span in Matrix Space)**.** In $\mathbb{R}^{2\times 2}$:

$$\text{span}\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \mathbb{R}^{2\times 2} \tag{57}$$

Any 2×2 matrix can be written as:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \tag{58}$$

3.3 Spanning Sets and Generating Sets

Definition 3.3 (Spanning Set)**.** A set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ **spans** a vector space V if:

$$V = \text{span}(S) \tag{59}$$

We say S is a **spanning set** or **generating set** for V.

Example 3.4 (Standard Spanning Sets)**. 1. Standard basis for $\mathbb{R}^3$:**

$$\mathbb{R}^3 = \text{span}\left\{ \begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix} \right\} \tag{60}$$

2. Alternative spanning set for $\mathbb{R}^2$:

$$\mathbb{R}^2 = \text{span}\left\{ \begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix} \right\} \tag{61}$$

Verify: Any $\begin{bmatrix}x\\y\end{bmatrix} = \alpha_1 \begin{bmatrix}1\\1\end{bmatrix} + \alpha_2 \begin{bmatrix}1\\-1\end{bmatrix}$
Solving: $\alpha_1 = \frac{x+y}{2}$, $\alpha_2 = \frac{x-y}{2}$
3. Polynomials P_2:

$$P_2 = \text{span}\{1, x, x^2\} \tag{62}$$

4 Linear Independence and Dependence

4.1 Definitions

Definition 4.1 (Linear Independence)**.** A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is **linearly independent** if the only solution to:

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = \mathbf{0} \tag{63}$$

is $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$ (the trivial solution).

Equivalently: No vector in the set can be written as a linear combination of the others.

Definition 4.2 (Linear Dependence)**.** A set of vectors is **linearly dependent** if it is not linearly independent. That is, there exist scalars $\alpha_1, \ldots, \alpha_n$ (not all zero) such that:

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = \mathbf{0} \tag{64}$$

Equivalently: At least one vector can be expressed as a linear combination of the others.

Example 4.1 (Testing Linear Independence in $\mathbb{R}^3$)**. Example 1:** Are $\mathbf{v}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$ linearly independent?

Setup:

$$\alpha_1 \begin{bmatrix}1\\0\\0\end{bmatrix} + \alpha_2 \begin{bmatrix}0\\1\\0\end{bmatrix} + \alpha_3 \begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \tag{65}$$

This gives:

$$\begin{bmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \tag{66}$$

Only solution: $\alpha_1 = \alpha_2 = \alpha_3 = 0$

Conclusion: Linearly independent

Example 2: Are $\mathbf{v}_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix}2\\4\\6\end{bmatrix}$ linearly independent?

Observation: $\mathbf{v}_2 = 2\mathbf{v}_1$

So: $2\mathbf{v}_1 - 1\mathbf{v}_2 = \mathbf{0}$ (non-trivial solution)

Conclusion: Linearly dependent

4.2 Testing Linear Independence

Theorem 4.1 (Matrix Method for Testing Independence)**.** Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ in $\mathbb{R}^m$ are linearly independent if and only if the matrix:

$$A = [\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_n] \tag{67}$$

has rank n (i.e., all columns are pivot columns).

Equivalently: The system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

Example 4.2 (Using Row Reduction)**.** Test if $\mathbf{v}_1 = \begin{bmatrix}1\\2\\1\end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix}2\\1\\0\end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix}1\\-1\\-1\end{bmatrix}$ are linearly independent.

Form matrix:

$$A = \begin{bmatrix}1 & 2 & 1\\2 & 1 & -1\\1 & 0 & -1\end{bmatrix} \tag{68}$$

Row reduce to RREF:

$$\begin{bmatrix}1 & 2 & 1\\2 & 1 & -1\\1 & 0 & -1\end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix}1 & 2 & 1\\0 & -3 & -3\\1 & 0 & -1\end{bmatrix} \tag{69}$$

$$\xrightarrow{R_3 - R_1} \begin{bmatrix}1 & 2 & 1\\0 & -3 & -3\\0 & -2 & -2\end{bmatrix} \tag{70}$$

$$\xrightarrow{R_2 \div (-3)} \begin{bmatrix}1 & 2 & 1\\0 & 1 & 1\\0 & -2 & -2\end{bmatrix} \tag{71}$$

$$\xrightarrow{R_3 + 2R_2} \begin{bmatrix}1 & 2 & 1\\0 & 1 & 1\\0 & 0 & 0\end{bmatrix} \tag{72}$$

Analysis: Only 2 pivot columns (rank = 2)
Conclusion: Linearly dependent (only 2 independent vectors among 3)
Find dependency: From RREF, x_3 is free. Let $x_3 = 1$:

$$x_2 + x_3 = 0 \Rightarrow x_2 = -1 \tag{73}$$

$$x_1 + 2x_2 + x_3 = 0 \Rightarrow x_1 = 1 \tag{74}$$

Dependency relation: $\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$
Or: $\mathbf{v}_3 = -\mathbf{v}_1 + \mathbf{v}_2$

4.3 Properties of Linear Independence

Theorem 4.2 (Key Properties)**.**

1. Any set containing the zero vector is linearly dependent
2. A set with exactly two vectors is linearly dependent if and only if one is a scalar multiple of the other
3. If $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent, then any subset is also linearly independent
4. If $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly dependent, then any superset is also linearly dependent
5. In $\mathbb{R}^n$, any set of more than n vectors is linearly dependent
6. A set of vectors is linearly dependent if and only if at least one vector is in the span of the others

Proof of Property 1. If $\mathbf{v}_1 = \mathbf{0}$, then:

$$1 \cdot \mathbf{0} + 0 \cdot \mathbf{v}_2 + \cdots + 0 \cdot \mathbf{v}_n = \mathbf{0} \tag{75}$$

This is a non-trivial linear combination (coefficient of $\mathbf{0}$ is $1 \neq 0$), so the set is linearly dependent. □

5 Basis and Dimension

5.1 Definition of Basis

Definition 5.1 (Basis)**.** A set $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a **basis** for vector space V if:

1. $\mathcal{B}$ is linearly independent
2. $\mathcal{B}$ spans V (i.e., $\text{span}(\mathcal{B}) = V$)

Equivalent definition: $\mathcal{B}$ is a basis if every vector in V can be expressed *uniquely* as a linear combination of vectors in $\mathcal{B}$.

Theorem 5.1 (Uniqueness of Representation)**.** If $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for V, then every $\mathbf{v} \in V$ has a unique representation:

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n \tag{76}$$

The coefficients $(\alpha_1, \alpha_2, \ldots, \alpha_n)$ are called the **coordinates** of $\mathbf{v}$ with respect to basis $\mathcal{B}$.

Proof. **Existence:** Since $\mathcal{B}$ spans V, every $\mathbf{v} \in V$ can be written as a linear combination.
Uniqueness: Suppose $\mathbf{v} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i = \sum_{i=1}^{n} \beta_i \mathbf{v}_i$
Then: $\sum_{i=1}^{n} (\alpha_i - \beta_i)\mathbf{v}_i = \mathbf{0}$
Since $\mathcal{B}$ is linearly independent: $\alpha_i - \beta_i = 0$ for all i
Therefore: $\alpha_i = \beta_i$ for all i □

5.2 Standard Bases

Example 5.1 (Standard Basis for $\mathbb{R}^n$)**.** The **standard basis** (or **canonical basis**) for $\mathbb{R}^n$ is:

$$\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\} \tag{77}$$

where:

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow i\text{-th position} \tag{78}$$

For $\mathbb{R}^3$:

$$\mathcal{E} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \tag{79}$$

Any vector $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$

Example 5.2 (Other Standard Bases)**. 1. Polynomial space P_n:**

$$\mathcal{B} = \{1, x, x^2, \ldots, x^n\} \tag{80}$$

Any polynomial: $p(x) = a_0(1) + a_1 x + a_2 x^2 + \cdots + a_n x^n$
2. Matrix space $\mathbb{R}^{2\times 2}$:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \tag{81}$$

3. Complex numbers $\mathbb{C}$ (as vector space over $\mathbb{R}$):

$$\mathcal{B} = \{1, i\} \tag{82}$$

Any complex number: $z = a(1) + b(i) = a + bi$

Example 5.3 (Non-Standard Bases for $\mathbb{R}^2$)**. Basis 1:**

$$\mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \tag{83}$$

Verify independence:

$$\alpha_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{84}$$

System: $\alpha_1 + \alpha_2 = 0$ and $\alpha_1 - \alpha_2 = 0$
Solution: $\alpha_1 = \alpha_2 = 0$ (linearly independent)
Verify spanning: Any $\begin{bmatrix} x \\ y \end{bmatrix}$ can be written as:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{x+y}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{x-y}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{85}$$

(spans $\mathbb{R}^2$)
Basis 2:

$$\mathcal{B}_2 = \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\} \tag{86}$$

Can verify similarly that this is also a basis for $\mathbb{R}^2$.

5.3 Dimension

Theorem 5.2 (Invariance of Basis Size)**.** If V has a finite basis, then all bases of V have the same number of elements.

Definition 5.2 (Dimension)**.** The **dimension** of a vector space V, denoted $\dim(V)$, is the number of vectors in any basis for V.
Convention: $\dim(\{\mathbf{0}\}) = 0$ (the zero space)
If V has no finite basis, we say V is **infinite-dimensional**.

Example 5.4 (Dimensions of Standard Spaces)**.**

$$\dim(\mathbb{R}^n) = n \tag{87}$$
$$\dim(\mathbb{C}^n) = n \quad (\text{over } \mathbb{C}), \quad 2n \quad (\text{over } \mathbb{R}) \tag{88}$$
$$\dim(P_n) = n + 1 \tag{89}$$
$$\dim(\mathbb{R}^{m \times n}) = mn \tag{90}$$
$$\dim(C[a,b]) = \infty \tag{91}$$

Theorem 5.3 (Dimension Theorems)**.** Let V be a vector space with $\dim(V) = n$.

1. Any linearly independent set of n vectors is a basis
2. Any spanning set of n vectors is a basis
3. Any linearly independent set can be extended to a basis
4. Any spanning set contains a basis
5. Any linearly independent set of more than n vectors in V is impossible
6. Any set of fewer than n vectors cannot span V

Theorem 5.4 (Rank-Nullity Theorem)**.** For a linear transformation $T : V \to W$ (or matrix $A \in \mathbb{R}^{m\times n}$):

$$\dim(V) = \dim(\ker(T)) + \dim(\text{im}(T)) \tag{92}$$

For matrices:

$$n = \dim(\mathcal{N}(A)) + \text{rank}(A) \tag{93}$$

where:

- $\mathcal{N}(A)$ = null space (kernel)
- $\text{rank}(A)$ = dimension of column space

Example 5.5 (Applying Rank-Nullity)**.** Consider $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

Row reduce:

$$\text{RREF}(A) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \tag{94}$$

Rank: 2 (two pivot columns)
Nullity: $\dim(\mathcal{N}(A)) = 3 - 2 = 1$
Null space basis: Solve $A\mathbf{x} = \mathbf{0}$:

$$\mathcal{N}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\} \tag{95}$$

Verify: $\dim(\mathcal{N}(A)) + \text{rank}(A) = 1 + 2 = 3 = n$

5.4 Subspace Dimensions

Theorem 5.5 (Dimension of Subspaces)**.** If W is a subspace of finite-dimensional vector space V, then:

1. $\dim(W) \leq \dim(V)$
2. $\dim(W) = \dim(V)$ if and only if $W = V$

Theorem 5.6 (Dimension of Sum)**.** For subspaces W_1 and W_2 of V:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) \tag{96}$$

Special case (Direct Sum): If $W_1 \cap W_2 = \{\mathbf{0}\}$:

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2) \tag{97}$$

6 Change of Basis

6.1 Coordinate Vectors

Definition 6.1 (Coordinate Vector)**.** Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis for V. For $\mathbf{v} \in V$ with:

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_n \mathbf{v}_n \tag{98}$$

The **coordinate vector** of $\mathbf{v}$ relative to $\mathcal{B}$ is:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} \tag{99}$$

Example 6.1 (Coordinate Vectors in $\mathbb{R}^2$)**.** Let $\mathbf{v} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ and two bases:

Standard basis: $\mathcal{E} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

Coordinates:

$$[\mathbf{v}]_{\mathcal{E}} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \tag{100}$$

Alternative basis: $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

Find α_1, α_2 such that:

$$\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{101}$$

System:

$$\alpha_1 + \alpha_2 = 5 \tag{102}$$
$$\alpha_1 - \alpha_2 = 3 \tag{103}$$

Solution: $\alpha_1 = 4$, $\alpha_2 = 1$

Coordinates:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \tag{104}$$

6.2 Change of Basis Matrix

Definition 6.2 (Change of Basis Matrix)**.** Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $\mathcal{C} = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ be two bases for V.

The **change of basis matrix** from $\mathcal{B}$ to $\mathcal{C}$ is:

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} [\mathbf{v}_1]_{\mathcal{C}} & [\mathbf{v}_2]_{\mathcal{C}} & \cdots & [\mathbf{v}_n]_{\mathcal{C}} \end{bmatrix} \tag{105}$$

This matrix satisfies:

$$[\mathbf{v}]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{v}]_{\mathcal{B}} \tag{106}$$

Theorem 6.1 (Properties of Change of Basis Matrix)**.**

1. $P_{\mathcal{C} \leftarrow \mathcal{B}}$ is invertible
2. $(P_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} = P_{\mathcal{B} \leftarrow \mathcal{C}}$

3. $P_{\mathcal{B}\leftarrow\mathcal{B}} = I$ (identity)
4. $P_{\mathcal{C}\leftarrow\mathcal{A}} = P_{\mathcal{C}\leftarrow\mathcal{B}} P_{\mathcal{B}\leftarrow\mathcal{A}}$ (composition)

Example 6.2 (Computing Change of Basis Matrix)**.** In $\mathbb{R}^2$, let:

$$\mathcal{B} = \left\{ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}, \quad \mathcal{E} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \tag{107}$$

Find $P_{\mathcal{E}\leftarrow\mathcal{B}}$:
Since $\mathcal{E}$ is the standard basis:

$$[\mathbf{v}_1]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad [\mathbf{v}_2]_{\mathcal{E}} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{108}$$

Therefore:

$$P_{\mathcal{E}\leftarrow\mathcal{B}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{109}$$

Find $P_{\mathcal{B}\leftarrow\mathcal{E}}$:
We need to express standard basis vectors in terms of $\mathcal{B}$:
For $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{110}$$

Solving: $\alpha_1 = \frac{1}{2}$, $\alpha_2 = \frac{1}{2}$
For $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \beta_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{111}$$

Solving: $\beta_1 = \frac{1}{2}$, $\beta_2 = -\frac{1}{2}$
Therefore:

$$P_{\mathcal{B}\leftarrow\mathcal{E}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \tag{112}$$

Verify:

$$P_{\mathcal{E}\leftarrow\mathcal{B}} P_{\mathcal{B}\leftarrow\mathcal{E}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \tag{113}$$

6.3 Algorithm for Change of Basis

Algorithm: Finding Change of Basis Matrix from $\mathcal{B}$ to $\mathcal{E}$

Given: Basis $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ in $\mathbb{R}^n$
Goal: Find $P_{\mathcal{B}\leftarrow\mathcal{E}}$ (from standard basis to $\mathcal{B}$)
Method:

1. Form augmented matrix $[B \mid I]$ where $B = [\mathbf{v}_1 \mid \cdots \mid \mathbf{v}_n]$
2. Row reduce to $[I \mid P]$
3. The resulting $P = P_{\mathcal{B}\leftarrow\mathcal{E}} = B^{-1}$

Note: $P_{\mathcal{E}\leftarrow\mathcal{B}} = B$ (just the matrix of basis vectors)

7 Applications in Machine Learning and Deep Learning

7.1 Feature Spaces and Data Representation

ML Application: Feature Spaces

In machine learning, data points are represented as vectors in a feature space.
Example: Image Classification

- Grayscale image: 28×28 pixels $\rightarrow$ vector in $\mathbb{R}^{784}$
- Color image: $224 \times 224 \times 3 \rightarrow$ vector in $\mathbb{R}^{150528}$
- Feature space: $V = \mathbb{R}^d$ where d is number of features

Linear Independence in Features:

- Redundant features $\rightarrow$ linearly dependent
- Feature selection aims to find linearly independent subset
- Multicollinearity in regression: features are linearly dependent

7.2 Principal Component Analysis (PCA)

PCA as Change of Basis

Goal: Find optimal basis (principal components) for data
Mathematical Framework:

1. Data matrix $X \in \mathbb{R}^{n\times d}$ (n samples, d features)
2. Compute covariance matrix $\Sigma = \frac{1}{n}X^T X$
3. Find eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_d$ of Σ
4. These form a new orthonormal basis
5. Project data onto this basis: $Z = XP$ where $P = [\mathbf{v}_1 \mid \cdots \mid \mathbf{v}_k]$

Key Insight:

- Original basis: standard features
- New basis: principal components (directions of maximum variance)
- Change of basis matrix: eigenvector matrix

Example 7.1 (PCA Example)**.** Dataset in $\mathbb{R}^2$: $\{(2,1),(3,2),(4,3),(5,4)\}$

Step 1: Center data Mean: $\bar{\mathbf{x}} = \begin{bmatrix} 3.5 \\ 2.5 \end{bmatrix}$

Step 2: Covariance matrix

$$\Sigma = \begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix} \tag{114}$$

Step 3: Eigenvalues and eigenvectors

$$\lambda_1 = 3.34, \quad \mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{115}$$

$$\lambda_2 = 0, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{116}$$

Step 4: New basis

$$\mathcal{B}_{PCA} = \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \tag{117}$$

Interpretation: First principal component captures all variance; data lies on a line in $\mathbb{R}^2$.

7.3 Neural Network Weight Spaces

Parameter Space in Deep Learning

Neural Network Parameters as Vector Space
For a neural network with L layers:

- Weights and biases: $\theta = (W^{(1)}, b^{(1)}, \ldots, W^{(L)}, b^{(L)})$
- Flatten into single vector: $\theta \in \mathbb{R}^p$ where $p =$ total parameters
- Parameter space is a vector space

Optimization as Vector Space Operation:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t) \tag{118}$$

This is vector addition in parameter space!
Gradient Descent Properties:

- Gradients form a vector space
- Linear combinations of gradients (momentum, Adam)
- Subspaces of parameters (layer-wise training)

7.4 Kernel Methods and Infinite Dimensions

Infinite-Dimensional Feature Spaces

Kernel Trick: Implicitly map data to infinite-dimensional space
Example: RBF Kernel

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2}\right) \tag{119}$$

Maps to infinite-dimensional Hilbert space $\mathcal{H}$ where:

- Inner product defined by kernel
- Linear methods in $\mathcal{H} \to$ non-linear in original space

Reproducing Kernel Hilbert Space (RKHS):

- Function space with special properties
- Basis functions: $\{k(\cdot, \mathbf{x}_i)\}_{i=1}^n$
- Representer theorem: solution lies in finite-dimensional subspace

7.5 Word Embeddings

Example 7.2 (Word2Vec as Vector Space)**. Embedding space:** $V = \mathbb{R}^{300}$ (typical dimension)

Each word mapped to vector: $w \mapsto \mathbf{v}_w \in \mathbb{R}^{300}$
Linear Structure:

$$\mathbf{v}_{\text{king}} - \mathbf{v}_{\text{man}} + \mathbf{v}_{\text{woman}} \approx \mathbf{v}_{\text{queen}} \tag{120}$$

Vector space operations:

- Addition: semantic combination
- Subtraction: removing concepts
- Scalar multiplication: amplifying meaning
- Cosine similarity: measuring semantic closeness

Span interpretation:

- Learned embeddings span a subspace of $\mathbb{R}^{300}$
- Actual dimension (intrinsic) often much smaller
- Basis: latent semantic dimensions

8 Worked Examples

8.1 Complete Example 1: Verifying a Basis

Example 8.1 (Verification Exercise)**. Problem:** Verify that $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

Solution:

Step 1: Check linear independence

Form matrix and row reduce:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \tag{121}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 1 & 0 & 1 \end{bmatrix} \tag{122}$$

$$\xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & -2 & 1 \end{bmatrix} \tag{123}$$

$$\xrightarrow{R_3 - \frac{2}{3}R_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \tag{124}$$

Three pivot columns $\rightarrow$ rank $= 3 \rightarrow$ linearly independent

Step 2: Check spanning

Since we have 3 linearly independent vectors in $\mathbb{R}^3$, and $\dim(\mathbb{R}^3) = 3$, they must span $\mathbb{R}^3$

Conclusion: $\mathcal{B}$ is a basis for $\mathbb{R}^3$.

8.2 Complete Example 2: Finding Coordinates

Example 8.2 (Coordinate Calculation)**. Problem:** Find the coordinates of $\mathbf{v} = \begin{bmatrix} 7 \\ 5 \\ 3 \end{bmatrix}$ with respect to basis $\mathcal{B}$ from Example 1.

Solution:

We need to find $\alpha_1, \alpha_2, \alpha_3$ such that:

$$\begin{bmatrix} 7 \\ 5 \\ 3 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \tag{125}$$

Set up augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 2 & 1 & 1 & 5 \\ 1 & 0 & 1 & 3 \end{array}\right] \tag{126}$$

Row reduce:

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 2 & 1 & 1 & 5 \\ 1 & 0 & 1 & 3 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & -3 & 1 & -9 \\ 1 & 0 & 1 & 3 \end{array}\right] \tag{127}$$

$$\xrightarrow{R_3-R_1} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & -3 & 1 & -9 \\ 0 & -2 & 1 & -4 \end{array}\right] \tag{128}$$

$$\xrightarrow{R_2\div(-3)} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & 1 & -\frac{1}{3} & 3 \\ 0 & -2 & 1 & -4 \end{array}\right] \tag{129}$$

$$\xrightarrow{R_3+2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & 1 & -\frac{1}{3} & 3 \\ 0 & 0 & \frac{1}{3} & 2 \end{array}\right] \tag{130}$$

$$\xrightarrow{R_3\times 3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & 1 & -\frac{1}{3} & 3 \\ 0 & 0 & 1 & 6 \end{array}\right] \tag{131}$$

Back substitution:

$$\alpha_3 = 6 \tag{132}$$

$$\alpha_2 - \frac{1}{3}(6) = 3 \Rightarrow \alpha_2 = 5 \tag{133}$$

$$\alpha_1 + 2(5) = 7 \Rightarrow \alpha_1 = -3 \tag{134}$$

Answer:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} -3 \\ 5 \\ 6 \end{bmatrix} \tag{135}$$

Verification:

$$-3\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + 5\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 6\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \tag{136}$$

$$= \begin{bmatrix} -3 \\ -6 \\ -3 \end{bmatrix} + \begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 6 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \\ 3 \end{bmatrix} \tag{137}$$

8.3 Complete Example 3: Dimension of Subspace

Example 8.3 (Finding Basis and Dimension)**. Problem:** Find a basis and dimension for:

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \mathbb{R}^4 : x + y - z = 0, \quad 2x - y + w = 0 \right\} \tag{138}$$

Solution:

Step 1: Express constraints in matrix form

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 2 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{139}$$

Step 2: Row reduce

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 2 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -3 & 2 & 1 \end{bmatrix} \tag{140}$$

Step 3: Identify free variables
Pivot columns: 1, 2 $\rightarrow$ Basic variables: x, y
Free variables: z, w
Step 4: Express basic variables in terms of free
From row 2: $-3y + 2z + w = 0 \Rightarrow y = \frac{2z+w}{3}$
From row 1: $x + \frac{2z+w}{3} - z = 0 \Rightarrow x = \frac{z-w}{3}$
Step 5: Parametric form

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} \frac{z-w}{3} \\ \frac{2z+w}{3} \\ z \\ w \end{bmatrix} = z \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ 1 \\ 0 \end{bmatrix} + w \begin{bmatrix} -\frac{1}{3} \\ \frac{1}{3} \\ 0 \\ 1 \end{bmatrix} \tag{141}$$

Step 6: Clear fractions (optional)
Multiply first vector by 3:

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 3 \end{bmatrix} \right\} \tag{142}$$

Answer:

- Basis: $\mathcal{B}_W = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 3 \end{bmatrix} \right\}$
- Dimension: $\dim(W) = 2$

Verify using Rank-Nullity:

$$\dim(W) = 4 - \text{rank}(A) = 4 - 2 = 2 \tag{143}$$

9 Assessment Questions

9.1 Level 1: Conceptual Understanding

Q1. Define a vector space and list the ten axioms.

Q2. What is the difference between a subspace and a subset?

Q3. Give three examples of vector spaces, one of which is not $\mathbb{R}^n$.
Q4. What does it mean for vectors to be linearly independent?
Q5. Define a basis and explain why the representation in a basis is unique.
Q6. What is the dimension of $\mathbb{R}^5$? What is the dimension of P_3?
Q7. True or False: Every linearly independent set is a basis. Explain.
Q8. True or False: The union of two subspaces is a subspace. Explain.
Q9. What is the zero vector in the vector space of 2×2 matrices?
Q10. Can a set of 4 vectors span $\mathbb{R}^5$? Explain.

9.2 Level 2: Computational

Q11. Determine if $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : 2x + 3y = 0 \right\}$ is a subspace of $\mathbb{R}^2$.

Q12. Test if $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \right\}$ is linearly independent.

Q13. Find a basis for the null space of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \end{bmatrix}$.

Q14. Express $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$ in terms of basis $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}$.

Q15. Find the dimension of span $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}$.

Q16. Compute the change of basis matrix from $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$ to the standard basis.

Q17. Find a basis for P_2 (polynomials of degree ≤ 2) different from $\{1, x, x^2\}$.
Q18. Is $\{1, \sin(x), \cos(x)\}$ linearly independent in $C[0, 2\pi]$?
Q19. Find the dimension of the intersection $W_1 \cap W_2$ where:

$$W_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} \tag{144}$$

$$W_2 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y + z = 0 \right\} \tag{145}$$

Q20. Extend $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ to a basis for $\mathbb{R}^3$.

9.3 Level 3: Advanced and Proofs

Q21. Prove that the intersection of two subspaces is a subspace.

Q22. Prove that if $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly dependent, then one vector is in the span of the others.

Q23. Prove the Rank-Nullity Theorem for a 3×4 matrix.
Q24. Show that $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$.

Q25. Prove that any $n+1$ vectors in $\mathbb{R}^n$ are linearly dependent.

Q26. Let V be finite-dimensional. Prove that any linearly independent set can be extended to a basis.

Q27. Show that the set of solutions to $A\mathbf{x} = \mathbf{0}$ forms a subspace.

Q28. Prove that if $\dim(V) = n$ and $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ spans V, then it is a basis.

Q29. Let $T : V \to W$ be linear. Prove that $\ker(T)$ is a subspace of V and $\text{im}(T)$ is a subspace of W.

Q30. Prove that for finite-dimensional spaces, $\dim(V^*) = \dim(V)$ where V^* is the dual space.

10 Solutions to Selected Problems

10.1 Solution to Q11

Problem: Determine if $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : 2x + 3y = 0 \right\}$ is a subspace of $\mathbb{R}^2$.

Solution:

Check 1: Contains zero

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} : \quad 2(0) + 3(0) = 0 \tag{146}$$

Check 2: Closed under addition

Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in W$

Then: $2u_1 + 3u_2 = 0$ and $2v_1 + 3v_2 = 0$

For $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$:

$$2(u_1 + v_1) + 3(u_2 + v_2) = 2u_1 + 2v_1 + 3u_2 + 3v_2 \tag{147}$$
$$= (2u_1 + 3u_2) + (2v_1 + 3v_2) \tag{148}$$
$$= 0 + 0 = 0 \tag{149}$$

Check 3: Closed under scalar multiplication

Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in W$ and $c \in \mathbb{R}$

Then: $2u_1 + 3u_2 = 0$

For $c\mathbf{u} = \begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}$:

$$2(cu_1) + 3(cu_2) = c(2u_1 + 3u_2) \tag{150}$$
$$= c(0) = 0 \tag{151}$$

Conclusion: W is a subspace of $\mathbb{R}^2$

Geometric interpretation: W is a line through the origin with slope $-\frac{2}{3}$.

10.2 Solution to Q12

Problem: Test if $\left\{ \begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}, \begin{bmatrix}7\\8\\9\end{bmatrix} \right\}$ is linearly independent.

Solution:
Form matrix and row reduce:

$$A = \begin{bmatrix}1 & 4 & 7\\2 & 5 & 8\\3 & 6 & 9\end{bmatrix} \tag{152}$$

$$\begin{bmatrix}1 & 4 & 7\\2 & 5 & 8\\3 & 6 & 9\end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix}1 & 4 & 7\\0 & -3 & -6\\3 & 6 & 9\end{bmatrix} \tag{153}$$

$$\xrightarrow{R_3-3R_1} \begin{bmatrix}1 & 4 & 7\\0 & -3 & -6\\0 & -6 & -12\end{bmatrix} \tag{154}$$

$$\xrightarrow{R_3-2R_2} \begin{bmatrix}1 & 4 & 7\\0 & -3 & -6\\0 & 0 & 0\end{bmatrix} \tag{155}$$

Analysis: Only 2 pivot columns $\rightarrow$ rank $= 2$
Conclusion: Linearly dependent
Dependency relation: From RREF, let $x_3 = 1$ (free variable):

$$-3x_2 - 6(1) = 0 \Rightarrow x_2 = -2 \tag{156}$$

$$x_1 + 4(-2) + 7(1) = 0 \Rightarrow x_1 = 1 \tag{157}$$

Therefore: $\begin{bmatrix}1\\2\\3\end{bmatrix} - 2\begin{bmatrix}4\\5\\6\end{bmatrix} + \begin{bmatrix}7\\8\\9\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$

10.3 Solution to Q13

Problem: Find a basis for the null space of $A = \begin{bmatrix}1 & 2 & 3 & 4\\2 & 4 & 6 & 8\end{bmatrix}$.

Solution:
Step 1: Row reduce

$$\begin{bmatrix}1 & 2 & 3 & 4\\2 & 4 & 6 & 8\end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix}1 & 2 & 3 & 4\\0 & 0 & 0 & 0\end{bmatrix} \tag{158}$$

Step 2: Identify pivot and free variables

- Pivot column: 1 $\rightarrow$ Basic variable: x_1
- Free variables: x_2, x_3, x_4

Step 3: Express basic in terms of free
From $x_1 + 2x_2 + 3x_3 + 4x_4 = 0$:

$$x_1 = -2x_2 - 3x_3 - 4x_4 \tag{159}$$

Step 4: Parametric solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2x_2 - 3x_3 - 4x_4 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \tag{160}$$

$$= x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \tag{161}$$

Answer:

$$\mathcal{B}_{\mathcal{N}(A)} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \tag{162}$$

Dimension: $\dim(\mathcal{N}(A)) = 3$
Verify Rank-Nullity: $4 = 1 + 3$

10.4 Solution to Q21 (Proof)

Theorem: If W_1 and W_2 are subspaces of V, then $W_1 \cap W_2$ is also a subspace of V.

Proof:
We must verify the three subspace conditions:
(1) Contains zero vector
Since W_1 is a subspace, $\mathbf{0} \in W_1$.
Since W_2 is a subspace, $\mathbf{0} \in W_2$.
Therefore, $\mathbf{0} \in W_1 \cap W_2$.
(2) Closed under addition
Let $\mathbf{u}, \mathbf{v} \in W_1 \cap W_2$.
Then $\mathbf{u}, \mathbf{v} \in W_1$ and $\mathbf{u}, \mathbf{v} \in W_2$.
Since W_1 is a subspace: $\mathbf{u} + \mathbf{v} \in W_1$.
Since W_2 is a subspace: $\mathbf{u} + \mathbf{v} \in W_2$.
Therefore, $\mathbf{u} + \mathbf{v} \in W_1 \cap W_2$.
(3) Closed under scalar multiplication
Let $\mathbf{v} \in W_1 \cap W_2$ and $c \in \mathbb{F}$.
Then $\mathbf{v} \in W_1$ and $\mathbf{v} \in W_2$.
Since W_1 is a subspace: $c\mathbf{v} \in W_1$.
Since W_2 is a subspace: $c\mathbf{v} \in W_2$.
Therefore, $c\mathbf{v} \in W_1 \cap W_2$.
All three conditions are satisfied, so $W_1 \cap W_2$ is a subspace. $\square$

11 Practice Problems

11.1 Computational Practice

P1. Determine which of the following are subspaces of $\mathbb{R}^3$:

(a) $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x = 0 \right\}$

(b) $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y = 1 \right\}$

(c) $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : xy = 0 \right\}$

(d) $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x = y = z \right\}$

P2. Find all values of k for which the vectors are linearly dependent:

$$\begin{bmatrix} 1 \\ k \\ 2 \end{bmatrix}, \begin{bmatrix} k \\ 1 \\ k \end{bmatrix}, \begin{bmatrix} 2 \\ k \\ 1 \end{bmatrix} \tag{163}$$

P3. Find a basis for the row space, column space, and null space of:

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 3 & 1 \\ 3 & 6 & 4 & 1 \end{bmatrix} \tag{164}$$

P4. Given basis $\mathcal{B} = \{1+x, 1-x, x^2\}$ for P_2, find the coordinates of $p(x) = 3+2x+x^2$.

P5. Extend $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ to a basis for $\mathbb{R}^4$.

11.2 Conceptual Practice

P6. Explain why the set of invertible $n \times n$ matrices is NOT a vector space.

P7. Give an example showing that the union of two subspaces is not necessarily a subspace.

P8. Why does every vector space have at least two subspaces?

P9. If $\dim(V) = n$, what is the maximum dimension of a proper subspace?

P10. Can a vector space have exactly 5 elements? Explain.

11.3 ML/DL Application Practice

P11. In neural network training, parameters form a vector space. If a network has:

- Layer 1: 100×784 weight matrix + 100 biases
- Layer 2: 10×100 weight matrix + 10 biases

What is the dimension of the parameter space?

P12. In PCA, you have data in $\mathbb{R}^{50}$ and want to reduce to 5 dimensions. What is the dimension of:

(a) The original feature space?

(b) The subspace spanned by principal components?

(c) The "lost" information subspace?

P13. Word embeddings map vocabulary of size 10,000 to $\mathbb{R}^{300}$. If the actual embedded vectors span a subspace of dimension 150:

(a) What is the intrinsic dimension of the embedding space?

(b) What does this suggest about the redundancy?

(c) How many basis vectors truly characterize the embeddings?

12 Summary and Key Takeaways

Essential Concepts

Vector Spaces:

- Abstract structure generalizing $\mathbb{R}^n$
- Defined by 10 axioms (5 for addition, 5 for scalar multiplication)
- Examples: $\mathbb{R}^n$, matrices, polynomials, functions

Subspaces:

- Subset that is itself a vector space
- Must contain $\mathbf{0}$, closed under addition and scalar multiplication
- Intersection of subspaces is a subspace
- Union generally is NOT a subspace

Linear Independence:

- Only trivial combination gives zero
- Test via row reduction (check for pivot columns)
- At most n independent vectors in $\mathbb{R}^n$

Basis and Dimension:

- Basis = linearly independent spanning set
- Unique representation in a basis
- Dimension = number of basis vectors
- All bases have same size

Key Theorems:

- Rank-Nullity: $n = \dim(\mathcal{N}(A)) + \text{rank}(A)$
- Dimension Formula: $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$
- Extension Theorem: Independent sets can be extended to bases
- Reduction Theorem: Spanning sets contain bases

ML/DL Connections

Feature Spaces:

- Data points are vectors in $\mathbb{R}^d$
- Linearly dependent features $\rightarrow$ redundancy
- Dimension reduction finds optimal subspaces

Parameter Spaces:

- Neural network weights form high-dimensional vector spaces
- Optimization moves through this space
- Gradients are vectors in parameter space

Transformations:

- PCA: change of basis to principal components
- Embeddings: mapping to learned vector spaces
- Kernels: implicit infinite-dimensional spaces

Practical Importance:

- Understanding dimensions $\rightarrow$ model complexity
- Basis choice $\rightarrow$ computational efficiency
- Subspace structure $\rightarrow$ regularization strategies

13 Further Reading and Resources

13.1 Textbooks

1. **Linear Algebra and Its Applications** by Gilbert Strang
   - Comprehensive coverage of vector spaces
   - Excellent geometric intuition
   - MIT OpenCourseWare lectures available
2. **Linear Algebra Done Right** by Sheldon Axler
   - Abstract, proof-based approach
   - Emphasizes vector spaces over matrices
   - Great for theoretical understanding
3. **Introduction to Linear Algebra** by Gilbert Strang
   - More accessible than "Applications"
   - Strong on applications
   - Good for self-study
4. **Matrix Analysis and Applied Linear Algebra** by Carl Meyer
   - Balance of theory and computation
   - Many worked examples
   - Free online version available
5. **Deep Learning** by Goodfellow, Bengio, and Courville
   - Chapter 2: Linear Algebra for ML/DL
   - Focus on practical applications
   - Connection to deep learning

13.2 Online Resources

1. **Khan Academy: Linear Algebra**
   - Free video lectures
   - Interactive exercises
   - `https://www.khanacademy.org/math/linear-algebra`
2. **3Blue1Brown: Essence of Linear Algebra**
   - Outstanding visual explanations
   - Geometric intuition
   - `https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab`

3. **MIT OCW 18.06: Linear Algebra**
    - Professor Gilbert Strang
    - Complete course materials
    - `https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/`
4. **Computational Linear Algebra for Coders**
    - Fast.ai course
    - Practical, code-first approach
    - `https://github.com/fastai/numerical-linear-algebra`

13.3 Research Papers and Articles

1. **"A Tutorial on Principal Component Analysis"** by Jonathon Shlens
    - Clear explanation of PCA
    - Connection to eigenspaces
2. **"The Matrix Calculus You Need For Deep Learning"** by Terence Parr and Jeremy Howard
    - Practical guide for ML practitioners
    - Vector space perspective
3. **"Understanding the difficulty of training deep feedforward neural networks"** by Glorot and Bengio
    - Initialization in parameter space
    - Geometric perspective

13.4 Interactive Tools

1. **GeoGebra**
    - Visualize vector spaces and transformations
    - `https://www.geogebra.org/`
2. **Desmos Matrix Calculator**
    - Interactive matrix operations
    - `https://www.desmos.com/matrix`
3. **Seeing Theory: Linear Algebra**
    - Visual, interactive explanations
    - `https://seeing-theory.brown.edu/`

14 Appendix: Python Implementations

14.1 Testing Linear Independence

```
import numpy as np

def is_linearly_independent(vectors):
    """
    Test if a set of vectors is linearly independent.

    Args:
        vectors: list of numpy arrays or 2D array (vectors as columns)

    Returns:
        bool: True if linearly independent, False otherwise
    """
    # Convert to matrix (vectors as columns)
    if isinstance(vectors, list):
        A = np.column_stack(vectors)
    else:
        A = vectors

    # Check rank
    rank = np.linalg.matrix_rank(A)
    num_vectors = A.shape[1]

    return rank == num_vectors

# Example usage
v1 = np.array([1, 2, 3])
v2 = np.array([4, 5, 6])
v3 = np.array([7, 8, 9])

vectors = [v1, v2, v3]
print(f"Linearly independent: {is_linearly_independent(vectors)}")
# Output: False (these are linearly dependent)

# Test with independent vectors
u1 = np.array([1, 0, 0])
u2 = np.array([0, 1, 0])
u3 = np.array([0, 0, 1])

print(f"Linearly independent: {is_linearly_independent([u1, u2, u3])}")
# Output: True
```

14.2 Finding Basis for Null Space

```
import numpy as np
from scipy.linalg import null_space
```

```
def find_null_space_basis(A, tol=1e-10):
    """
    Find a basis for the null space of matrix A.

    Args:
        A: numpy array (m x n matrix)
        tol: tolerance for zero singular values

    Returns:
        basis: numpy array with basis vectors as columns
    """
    # Use SVD-based null space computation
    ns = null_space(A, rcond=tol)

    # If null space is trivial
    if ns.shape[1] == 0:
        print("Null space contains only zero vector")
        return None

    return ns

# Example
A = np.array([[1, 2, 3, 4],
              [2, 4, 6, 8]])

basis = find_null_space_basis(A)
print("Null space basis:")
print(basis)

# Verify: A @ basis should be approximately zero
print("\nVerification (should be near zero):")
print(A @ basis)
```

14.3 Coordinate Vector Computation

```
import numpy as np

def find_coordinates(v, basis):
    """
    Find coordinates of vector v with respect to given basis.

    Args:
        v: numpy array (vector to represent)
        basis: list of numpy arrays or 2D array (basis vectors as columns)

    Returns:
        coords: numpy array (coordinate vector)
```

```
    """
    # Form basis matrix
    if isinstance(basis, list):
        B = np.column_stack(basis)
    else:
        B = basis

    # Solve B @ coords = v
    coords = np.linalg.solve(B, v)

    return coords

# Example in R^2
v = np.array([5, 3])

# Standard basis
e1 = np.array([1, 0])
e2 = np.array([0, 1])
coords_std = find_coordinates(v, [e1, e2])
print(f"Coordinates in standard basis: {coords_std}")

# Alternative basis
b1 = np.array([1, 1])
b2 = np.array([1, -1])
coords_alt = find_coordinates(v, [b1, b2])
print(f"Coordinates in alternative basis: {coords_alt}")

# Verify
reconstruction = coords_alt[0] * b1 + coords_alt[1] * b2
print(f"Reconstruction: {reconstruction}")
print(f"Original: {v}")
```

14.4 Change of Basis Matrix

```
import numpy as np

def change_of_basis_matrix(old_basis, new_basis):
    """
    Compute change of basis matrix from old_basis to new_basis.

    Args:
        old_basis: list of vectors or matrix (old basis vectors as columns)
        new_basis: list of vectors or matrix (new basis vectors as columns)

    Returns:
        P: change of basis matrix such that [v]_new = P @ [v]_old
    """
    # Form basis matrices
```

```
    if isinstance(old_basis, list):
        B_old = np.column_stack(old_basis)
    else:
        B_old = old_basis

    if isinstance(new_basis, list):
        B_new = np.column_stack(new_basis)
    else:
        B_new = new_basis

    # P = B_new^(-1) @ B_old
    P = np.linalg.solve(B_new, B_old)

    return P

# Example
# Old basis
b1_old = np.array([1, 1])
b2_old = np.array([1, -1])

# New basis (standard)
b1_new = np.array([1, 0])
b2_new = np.array([0, 1])

P = change_of_basis_matrix([b1_old, b2_old], [b1_new, b2_new])
print("Change of basis matrix:")
print(P)

# Test with a vector
v_old_coords = np.array([4, 1])  # coordinates in old basis
v_new_coords = P @ v_old_coords   # coordinates in new basis
print(f"\nCoordinates in old basis: {v_old_coords}")
print(f"Coordinates in new basis: {v_new_coords}")
```

14.5 Dimension of Subspace Intersection

```
import numpy as np

def subspace_intersection_dim(A, B):
    """
    Find dimension of intersection of two subspaces.

    Args:
        A: matrix whose columns span subspace W1
        B: matrix whose columns span subspace W2

    Returns:
        dim: dimension of W1   W2
```

```
    """
    # Combine into one matrix
    AB = np.hstack([A, B])

    # Find rank of combined space (W1 + W2)
    rank_sum = np.linalg.matrix_rank(AB)

    # Find individual ranks
    rank_A = np.linalg.matrix_rank(A)
    rank_B = np.linalg.matrix_rank(B)

    # Use dimension formula:
    # dim(W1 + W2) = dim(W1) + dim(W2) - dim(W1   W2)
    dim_intersection = rank_A + rank_B - rank_sum

    return dim_intersection

# Example
# W1 spanned by columns of A
A = np.array([[1, 0],
              [0, 1],
              [1, 1]])

# W2 spanned by columns of B
B = np.array([[1, 1],
              [1, 1],
              [0, 2]])

dim = subspace_intersection_dim(A, B)
print(f"Dimension of intersection: {dim}")
```

15 Conclusion

Vector spaces form the mathematical foundation for understanding linear algebra and its applications in machine learning and deep learning. The concepts covered in these notes—from basic definitions to advanced topics like change of basis—are essential for:

- Understanding how data is represented and transformed
- Designing efficient neural network architectures
- Implementing dimensionality reduction techniques
- Analyzing optimization algorithms
- Developing new machine learning methods

Mastery of vector spaces provides the geometric and algebraic intuition needed to reason about high-dimensional spaces, linear transformations, and the fundamental structures underlying modern AI systems.

Key recommendations for continued learning:

1. Practice computational skills through problem-solving
2. Develop geometric intuition through visualization
3. Connect abstract concepts to ML/DL applications
4. Implement algorithms to solidify understanding
5. Study proofs to deepen theoretical knowledge

The journey from scalars to vectors to general vector spaces reveals the beautiful unity of mathematical structures. These same principles that govern simple 2D and 3D spaces extend seamlessly to the million-dimensional parameter spaces of modern deep neural networks.

"In mathematics, you don't understand things. You just get used to them."
— John von Neumann